

# Pramo Jewels

## AI Database Guide

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| Field    | Value                                                                                     |
|----------|-------------------------------------------------------------------------------------------|
| Project  | Pramo Jewels Premium E-Commerce Platform                                                  |
| Purpose  | Define database development standards for AI-generated schemas, queries, migrations, etc. |
| Audience | Database Engineers, Backend Developers, AI Coding Assistants, Architects, QA Engineers    |
| Version  | 1.0                                                                                       |

## AI Database Development Standards

| Category      | Guidelines                                                 |
|---------------|------------------------------------------------------------|
| Schema Design | Normalized tables with clear relationships and constraints |
| Naming        | Consistent table, column, index and constraint names       |
| Migrations    | Version-controlled, reversible and idempotent migrations   |
| Queries       | Parameterized, optimized and indexed SQL/ORM queries       |
| Transactions  | Use ACID transactions for critical operations              |
| Security      | Least privilege, encryption and input validation           |
| Backups       | Automated backups with tested restoration procedures       |
| Testing       | Migration, integrity and performance validation            |

## Objectives

Ensure AI-generated database changes preserve data integrity, performance, scalability and maintainability.

## Schema Standards

Define primary keys, foreign keys, indexes, unique constraints and audit fields consistently across all entities.

## Query Guidelines

Avoid N+1 queries, optimize joins, paginate large datasets and use prepared statements or ORM parameter binding.

## Migration Rules

Every schema change must include rollback support, compatibility checks and deployment validation.

## **Security Requirements**

Protect sensitive data through encryption where applicable, restrict database privileges and never expose credentials.

## **Performance Guidelines**

Review execution plans, optimize indexes, archive historical data and monitor slow queries.

## **Documentation**

Document ER changes, migration rationale, stored procedures, triggers and data dictionary updates.

## **Acceptance Criteria**

AI-generated database artifacts pass migration tests, maintain referential integrity, meet performance expectations and comply with project standards.